

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
saedabuzaytoun@saedabuzaytoun:~$ ls /etc/ > file.txt
saedabuzaytoun@saedabuzaytoun:~$ ls >> file.txt
saedabuzaytoun@saedabuzaytoun:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
```

Рисунок 1: Запись в файл

```
saedabuzaytoun@saedabuzaytoun:~$  
saedabuzaytoun@saedabuzaytoun:~$ grep .conf file.txt > conf.txt  
saedabuzaytoun@saedabuzaytoun:~$ cat conf.txt  
anthy-unicode.conf  
asound.conf  
brltty.conf  
chrony.conf  
cpupower-service.conf  
dconf  
dnsmasq.conf  
dracut.conf  
dracut.conf.d  
fprntd.conf  
fuse.conf  
host.conf  
idmapd.conf  
kdump.conf  
krb5.conf  
krb5.conf.d  
ld.so.conf  
ld.so.conf.d  
libaudit.conf  
libuser.conf  
locale.conf
```

Рисунок 2: Поиск расширения .conf


```
/home/saedabuzaytoun/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage03/report/bib/cite.bib
/home/saedabuzaytoun/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage04/report/_resources/csl
/home/saedabuzaytoun/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage04/report/bib/cite.bib
/home/saedabuzaytoun/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage05/report/_resources/csl
/home/saedabuzaytoun/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage05/report/bib/cite.bib
/home/saedabuzaytoun/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage06/report/_resources/csl
/home/saedabuzaytoun/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/project-personal/stage06/report/bib/cite.bib
/home/saedabuzaytoun/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/_resources/csl
/home/saedabuzaytoun/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/presentation/report/bib/cite.bib
/home/saedabuzaytoun/git-extended/.git/hooks/commit-msg.sample
/home/saedabuzaytoun/git-extended/.git/objects/87/cda5f005b32b5dcdb029c6707a35359f742c68
/home/saedabuzaytoun/git-extended/.git/config
/home/saedabuzaytoun/conf.txt
saedabuzaytoun@saedabuzaytoun:~$
```

Рисунок 3: Поиск файлов

```
/etc/foomatic/hashe.s.d/hashe.s.new
find: '/etc/libvirt': Permission denied
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
find: '/etc/lvm/archive': Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/backup': Permission denied
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
:find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss.d': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
saedabuzaytoun@saedabuzaytoun:~$  
saedabuzaytoun@saedabuzaytoun:~$ find /etc -name "h*" -print | less  
find: '/etc/audit': Permission denied  
saedabuzaytoun@saedabuzaytoun:~$  
saedabuzaytoun@saedabuzaytoun:~$ find ~ -name "log*" > logfile &  
[1] 6398  
saedabuzaytoun@saedabuzaytoun:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
saedabuzaytoun@saedabuzaytoun:~$ rm logfile  
saedabuzaytoun@saedabuzaytoun:~$
```

Рисунок 5: Фоновый запуск процесса

```
saedabuzaytoun@saedabuzaytoun:~$  
saedabuzaytoun@saedabuzaytoun:~$ gedit &  
[1] 6442  
saedabuzaytoun@saedabuzaytoun:~$  
saedabuzaytoun@saedabuzaytoun:~$ ps | grep gedit  
6442 pts/0    00:00:00 gedit  
saedabuzaytoun@saedabuzaytoun:~$ kill 6442  
[1]+  Terminated                  gedit  
saedabuzaytoun@saedabuzaytoun:~$  
saedabuzaytoun@saedabuzaytoun:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space available on the file system containing each file name argument.  If no file name is given, the space available on all currently mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file system, df shows the space available on that file system rather than on the file system containing the device node.  This version of df cannot show the space available on unmounted file systems, because on most kinds of systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
DU(1)                                User Commands                                DU(1)

NAME
    du - estimate file space usage

SYNOPSIS
    du [OPTION]... [FILE]...
    du [OPTION]... --files0-from=F

DESCRIPTION
    Summarize device usage of the set of FILES, recursively for directories.

    Mandatory arguments to long options are mandatory for short options too.

    -0, --null
        end each output line with NUL, not newline

    -a, --all
        write counts for all files, not just directories

    --apparent-size
        print apparent sizes rather than device usage; although the apparent size is usually smaller, it may be larger due to holes in ('sparse') files, internal fragmentation, indirect blocks, and the like

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

Manual page du(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
saedabuzaytoun@saedabuzaytoun:~$ df
Filesystem      1K-blocks    Used Available Use% Mounted on
/dev/nvme0n1p3 155186176 89712064 64809664  59% /
devtmpfs         4011284      0    4011284   0% /dev
tmpfs            4035660      92    4035568   1% /dev/shm
tmpfs            1614264     1952    1612312   1% /run
tmpfs             1024        0       1024    0% /run/credentials/systemd-journald.service
tmpfs            4035660     1608    4034052   1% /tmp
/dev/nvme0n1p3 155186176 89712064 64809664  59% /home
/dev/nvme0n1p2 1992552    518800    1352512  28% /boot
/dev/loop2        49280     49280        0 100% /var/lib/snapd/snap/snapd/25935
/dev/loop0        68480     68480        0 100% /var/lib/snapd/snap/core24/1349
/dev/loop1       115712    115712        0 100% /var/lib/snapd/snap/hugo/25676
tmpfs             1024        0       1024    0% /run/credentials/systemd-resolved.service
tmpfs            807132      168    806964   1% /run/user/1032
/dev/sr0         2677920    2677920        0 100% /run/media/saedabuzaytoun/Fedora-WS-Live-43
saedabuzaytoun@saedabuzaytoun:~$
```

Рисунок 9: Справка по команде du

```
4      ./git-extended/.git/objects/2b
4      ./git-extended/.git/objects/f9
4      ./git-extended/.git/objects/87
4      ./git-extended/.git/objects/db
76     ./git-extended/.git/objects
8      ./git-extended/.git/logs/refs/heads
8      ./git-extended/.git/logs/refs/remotes/origin
8      ./git-extended/.git/logs/refs/remotes
16     ./git-extended/.git/logs/refs
20     ./git-extended/.git/logs
212    ./git-extended/.git
220    ./git-extended
0      ./monthly
0      ./reports/monthly/monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
813932 .
saedabuzaytoun@saedabuzaytoun:~$
```

Рисунок 10: Запуск команды du


```
/home/saedabuzaytoun/git-extended/.git/objects/d6
/home/saedabuzaytoun/git-extended/.git/objects/ab
/home/saedabuzaytoun/git-extended/.git/objects/22
/home/saedabuzaytoun/git-extended/.git/objects/3e
/home/saedabuzaytoun/git-extended/.git/objects/3b
/home/saedabuzaytoun/git-extended/.git/objects/2b
/home/saedabuzaytoun/git-extended/.git/objects/f9
/home/saedabuzaytoun/git-extended/.git/objects/87
/home/saedabuzaytoun/git-extended/.git/objects/db
/home/saedabuzaytoun/git-extended/.git/logs
/home/saedabuzaytoun/git-extended/.git/logs/refs
/home/saedabuzaytoun/git-extended/.git/logs/refs/heads
/home/saedabuzaytoun/git-extended/.git/logs/refs/remotes
/home/saedabuzaytoun/git-extended/.git/logs/refs/remotes/origin
/home/saedabuzaytoun/monthly
/home/saedabuzaytoun/reports
/home/saedabuzaytoun/reports/monthly
/home/saedabuzaytoun/reports/monthly/monthly
/home/saedabuzaytoun/ski.places
/home/saedabuzaytoun/ski.places/equipment
/home/saedabuzaytoun/ski.places/plans
/home/saedabuzaytoun/australia
/home/saedabuzaytoun/play
/home/saedabuzaytoun/play/games
/home/saedabuzaytoun/play/games/play
saedabuzaytoun@saedabuzaytoun:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.